

1. Track Selection and Problem Framing

We target **Track A: Financial Volatility Prediction** for one-day-ahead realized-variance forecasting and regime classification. Volatility has multi-scale nonlinear dynamics—short-lag decay coexisting with hyperbolic long-lag decay [1, 2]—that GARCH-family models handle poorly near transition windows, where errors cause mispricing, hedging overruns, and margin miscalibration. QRC is well-suited: (i) no gradient-based quantum training is required, sidestepping barren plateaus; (ii) fading memory matches long-range autocorrelation of $\log\text{-}RV$; and (iii) the 2^n -dimensional Hilbert space provides polynomial-order feature interactions classically hard to replicate [3, 4]. Prior QRC covers realized-volatility [4] and chaos [5], but lacks *explicit timescale separation*. HAR models [1] decompose volatility into ~ 1 -, ~ 5 -, and ~ 22 -day components; we import this structure through phase-separated Hamiltonian parametrisation.

2. QRC Architecture Design

2.1 Reservoir Hamiltonians

Both reservoirs share the transverse-field Ising structure

$$H^{(r)} = J^{(r)} \sum_{\langle i,j \rangle} Z_i Z_j + h^{(r)} \sum_i X_i, \quad r \in \{\text{short, long}\}, \quad (1)$$

but differ in coupling and depth. The *short-memory* reservoir ($J^s = 0.3$, $h^s = 1.0$, $p^s = 2$, $J/h = 0.3$) sits in the paramagnetic phase ($J < h$), giving fast decorrelation and shock sensitivity. The *long-memory* reservoir ($J^l = 1.2$, $h^l = 0.4$, $p^l = 6$, $J/h = 3.0$) operates above the ferromagnetic critical point ($J > h$), sustaining correlations for regime tracking. This two-phase design follows the memory–nonlinearity trade-off of Čindrak et al. [6]: a single reservoir cannot simultaneously maximise fading memory and instantaneous input sensitivity, a limitation of single-reservoir QRC.

2.2 Encoding, Readout, and Feedback

Input x_t ($\log\text{-}RV$, normalised to $[-\pi/2, \pi/2]$) is encoded via $|\psi_0\rangle \leftarrow \prod_i RY_i(x_t)|0\rangle$ before each Trotter layer (feature re-uploading [3]); Garman–Klass estimates, VIX–RV spread, and log-returns occupy dedicated qubit subsets. The readout extracts: (1) Pauli expectations $\langle Z_i \rangle$ and $\langle Z_i Z_j \rangle$ ($2n-1$ observables per reservoir); (2) entanglement-spectrum features $\xi_k = -\ln \lambda_k$ from the half-chain reduced density matrix [7], retaining $m_\xi = \min(2^{n/2}, 8)$ eigenvalues. The concatenated $\approx 4n + m_\xi$ features feed a ridge-regression head—no quantum gradients; learning is closed-form. Feedback preserves fading memory. Two regime-conditioned variants—*QRC Dual+Ent+Regime* (one-hot HMM label) and *QRC Dual+Ent+Regime** (additional in-sample regime-mean bias correction)—both use training-only labels (no leakage).

3. Theoretical Justification

Why quantum reservoirs? A transverse-field Ising chain explores a 2^n -dimensional Hilbert space, creating polynomial-order interactions hard to replicate [3]. For volatility’s non-Gaussian tails, heavy kurtosis, and regime-switching [2], this expansion addresses GARCH’s linear conditional-variance limitation. Unlike LSTM or ESN, the closed-form readout resists overfitting on scarce high-volatility samples.

Why dual timescales? HAR [1] decomposes realised variance into ~ 1 -, ~ 5 -, and ~ 22 -day components; a single reservoir conflates all three via one coupling regime. Parametrising $J^s \ll J^l$ maps short/long HAR components onto the paramagnetic (fast-relaxing) and ferromagnetic (slow-relaxing) phases, as predicted by Čindrak et al. [6]. Phase II confirms: dual architecture reduces Lorenz RMSE by 85.2% over the single long-memory reservoir, where fast/slow modes need separate couplings.

Why entanglement-spectrum features? Pauli expectations capture only single- and two-body correlations; entanglement energies ξ_k encode multi-body structure [7], sensitive to regime-transition manifold traversals. We claim no universal advantage, only incremental signal: Phase II shows entanglement augmentation improves financial QLIKE from 0.371 to 0.362 and gives -68.4% Lorenz RMSE.

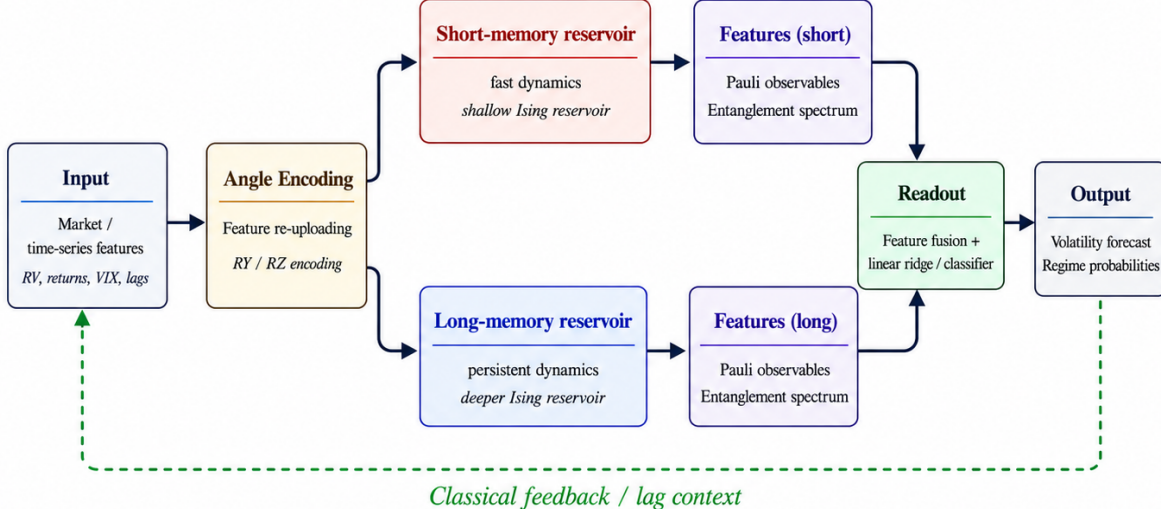


Figure 1: Dual-timescale QRC: market features $\rightarrow$ angle encoding $\rightarrow$ short ($J=0.3$, $p=2$) and long ($J=1.2$, $p=6$) Ising reservoirs $\rightarrow$ Pauli + entanglement features $\rightarrow$ ridge readout $\rightarrow$ forecast and regime probs.

4. Data Modeling Strategy

Datasets. (i) S&P 500 OHLCV (Yahoo Finance, ≈ 2008 – 2024 , 4 001 rows): Garman–Klass proxy used throughout (Oxford-Man RV unavailable; all models use identical data; 5-min RV replication is Phase 3). (ii) CBOE VIX (FRED, ≈ 2008 – 2024): VIX–RV spread feature and secondary target. (iii) Mackey–Glass / Lorenz (generated): chaotic architecture ablations. Financial splits: train = 3 200, test = 801 (≈ 2008 – 2020 / 2021 – 2024); chaotic: 7 991/1 998 (MG) and 3 991/998 (Lorenz), using identical splits (audit: all datasets `fair_split=True`). HMM thresholds: low/mid = 4.68×10^{-5} , mid/high = 7.85×10^{-5} ; test counts Low 517, Med 142, High 142.

Preprocessing. Log-transform and min-max normalise RV_t to $[-\pi/2, \pi/2]$; $L=22$ -day rolling window; target $\log RV_{t+1}$; scaler fit on training only. Classical models use log-space, inverse-transformed for metrics. HAR-RV coefficients: $\hat{\beta}_d=0.288$, $\hat{\beta}_w=0.479$, $\hat{\beta}_m=0.036$ (sum = 0.803). LSTM early-stopped epoch 21; XGBoost 97 rounds.

Baselines and metrics. GARCH(1,1) [8], HAR-RV [1], ARIMA(5,0,0), ESN [5], LSTM, and XGBoost. Metrics: RMSE, QLIKE [9], HMM regime accuracy, and volatility-targeting Sharpe (positions sized inversely to forecast variance).

5. Phase II Benchmark Results

5.1 Chaotic Time-Series: Mackey–Glass and Lorenz

Table 1: One-step-ahead RMSE, normalised MG and Lorenz ($n=9$ qubits; test 1 998/998).

Model	Src	MG RMSE	Lorenz RMSE
ESN	Classical	0.000158	0.000031
QRC Dual Pauli+Ent	QRC	0.006371	0.000542
QRC Dual Pauli	QRC	0.006680	0.001717
XGBoost	Classical	0.004007	0.003069
QRC Single Long Pauli+Ent	QRC	0.007308	0.003674
LSTM	Classical	0.005272	0.004444

On **Lorenz**, QRC Dual Pauli+Ent (RMSE 0.000542) ranks second, surpassing XGBoost by 82.3%, LSTM by 87.8%, and single-reservoir QRC by 85.2%, confirming timescale decomposition. Entanglement alone contributes -68.4% RMSE over Pauli-only. On **Mackey–Glass**, ESN dominates (RMSE 0.000158) because dense recurrence matches the series’ short-range autocorrelation; larger-qubit QRC for MG is a Phase 3

priority.

5.2 Financial Volatility: S&P 500 Realized Variance

Table 2: Phase II benchmark, S&P 500 RV (test 801 days), ranked by RMSE. *=regime-blend bias-correction. Oxford-Man RV identical (GK fallback).

Model	Src	RMSE	QLIKE	Reg.Acc.	Sharpe
QRC Dual+Ent+Regime	QRC	0.000107	0.3583	75.2%	1.257
QRC Dual+Ent	QRC	0.000108	0.3618	74.4%	1.197
QRC Dual	QRC	0.000112	0.3705	72.9%	1.112
LSTM-log	Classical	0.000117	0.4036	75.8%	1.297
QRC Dual+Ent Regime-Blend	QRC	0.000119	0.3508	74.3%	1.237
QRC Dual+Ent+Regime*	QRC	0.000119	0.2999	75.0%	1.249
ESN-log	Classical	0.000120	0.3752	73.4%	0.966
QRC Dual+Ent*	QRC	0.000120	0.3068	71.8%	1.045
XGBoost-log	Classical	0.000127	0.3791	73.0%	1.455
HAR-RV	Classical	0.000128	0.3486	72.4%	1.256
QRC Single+Ent	QRC	0.000132	0.3739	72.7%	1.128
ARIMA-log	Classical	0.000137	0.4196	73.3%	1.399
GARCH(1,1)	Classical	0.000156	0.4860	40.8%	0.543

RMSE. QRC Dual+Ent+Regime achieves RMSE 0.000107, an 8.5% gain over LSTM, 16.4% over HAR-RV, and 31.4% over GARCH(1,1). Ablation: Pauli-only dual $\rightarrow$ +Ent (-3.6%) $\rightarrow$ +Regime (-0.9%) isolates each component’s independent contribution. **QLIKE.** QRC Dual+Ent+Regime* achieves QLIKE 0.300, a 13.9% improvement over HAR-RV and 38.2% over GARCH—commercially important because QLIKE aligns with derivatives-pricing loss and variance-swap mispricing. **Regime accuracy and Sharpe.** LSTM leads narrowly (75.8% vs. QRC 75.2%); GARCH’s 40.8% is near-random. XGBoost leads Sharpe (1.455); QRC (1.257) matches HAR-RV (1.256) and Buy&Hold (1.258). The RMSE/QLIKE–Sharpe gap reflects tail sensitivity in the strategy; tail-aware readout calibration is a Phase 3 priority. **VIX.** QRC Single Long Pauli+Ent achieves best VIX RMSE (1.738) vs. ARIMA (1.742) and LSTM (1.743); the near-tie reflects near-unit-root dynamics with limited nonlinear headroom.

6. Quantum Platform and Resource Planning

Phase 2 uses **qBraid Lab** with **PennyLane Lightning.GPU** (NVIDIA cuQuantum) and Qiskit state-vector simulators at $n = 9$ qubits; results reproduce on CPU/GPU. A qBraid Skill and public GitHub repository accompany Phase 3. Phase 3 stages: **(1) Simulator sweep** over $n \in \{5, 8, 10, 12, 15\}$ and $p \in \{2, 4, 6\}$ Trotter steps (Lightning.GPU, feasible to $n = 20$ on A100), establishing RMSE/QLIKE scaling curves; **(2) Noise study** with calibrated depolarising/amplitude-damping noise plus Mitq ZNE, estimating fidelity needed to preserve QLIKE advantage; **(3) QPU validation** on `ibm_brisbane` via qBraid ($n = 8\text{--}12$, $p = 2$, 1 024–4 096 shots, shallow circuits prioritised). Depth $\mathcal{O}(p \cdot n)$; CNOT count $\approx 2pn$; shot budget $\approx 6.1 \times 10^6$ —feasible via batched qBraid SDK. Fallback: `qiskit_aer` calibrated noise models.

7. Stakeholder Impact and Phase 3 Plan

Beneficiaries. JonesTrading execution desks gain earlier regime-shift signals for TWAP-to-aggressive transitions, reducing shortfall. Risk managers gain 8.5%-lower-RMSE drawdown warnings for position sizing. Derivatives desks gain regime-aware Heston/SABR vol surfaces; the 13.9% QLIKE gain over HAR-RV reduces variance-swap mispricing directly. Regulatory teams gain an auditable ridge readout over fixed observables.

Phase 3 milestones. (1) Wk 1–2: replicate against Oxford-Man 5-min RV; clean RMSE/QLIKE table on authentic target. (2) Wk 3–4: simulator sweep (n, p^s, p^l); validate QLIKE scaling $n = 5\text{--}15$. (3) Wk 5–6: noise study + Mitq ZNE; QPU spot-check IBM via qBraid. (4) Wk 7–8: MNIST expressivity benchmark [3]; qBraid Skill + GitHub; Phase 3 paper with noise characterisation and hardware results.

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